

2.1 Algebra

Pure Mathematics 2 · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Binomial expansion for non-integer powers is infinite and valid only for $|x| < 1$ in $(1+x)^n$. Partial fractions simplify rational integration; first factor denominator and match numerator degrees. Factor theorem provides roots and polynomial division reduces degree.

Worked example: follow the reasoning

$(1+x)^{-1} = 1 - x + x^2 - \dots$ for $|x| < 1$, so $(1+2x)^{-1} \approx 1 - 2x + 4x^2$ when $|2x| < 1$.

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Expand $(1+x)^{-2}$ to terms in x^2 and state validity.

[3]

Final answer: _____

Answers and explanation

Question 1 [3 marks]

Use binomial coefficients: $1-2x+3x^2$ [2]; $\text{abs}(x)<1$ [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Mathematics 9709, 2026–2027. <https://www.cambridgeinternational.org/Images/697427-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.